GARAGEGHOST - COMPLETE ESP32-S3 VOICE ASSISTANT WIRING

A0 - all connections on one sheet

MAIN ENCLOSURE - COMMON LOW-VOLTAGE SYSTEM

5 V / 3 A PSU
isolated

+5V — W19 — P01 +5 V
GND — W24 — P02 GND

ESP32-S3-N16R8
main controller

5V IN — W19
GND — W24
GND-A — W02
3V3-A — W01
GPIO4 — W03
GPIO5 — W04
GPIO6 — W05
3V3-B
GPIO15 — W06
GPIO18 — W07
GPIO16 — W10
GPIO7 — W11
GPIO8 — W12
GPIO9 — W13
GPIO13 — W14
GPIO14 - NC
3V3-C — W32
GPIO10 — W29
GPIO11 — W30
GPIO12 — W31

REMOTE POD - CABLE <= 20 cm

INMP441
3.3 V only

GND — W02
C1 100 nF
VDD — W01
SCK — W03
WS — W04
SD — W05
L/R to GND
L/R

BUTTON
NO

R2 10 kohm
SW — W06
GND

LED

R3 1 kohm
A — W07
K

ASRPRO
offline wake-word recognition

R1 1 kohm
PA2 / TX1 — W10
5V — W20
GND — W25

MAX98357A
I2S amplifier

BCLK — W11
LRC — W12
DIN — W13
SD — W14
SPK+ — W15
SPK- — W16
GAIN - NC
VIN 5V
GND

SPEAKER
4 ohm / 3 W

\+ — W15
\- — W16

BSS138 4-channel
3.3 V <-> 5 V

LV 3V3 — W32
LV1 — W29
LV2 — W30
LV3 — W31
LV4 - UNUSED
HV1 — W34 — IN1
HV2 — W35 — IN2
HV3 — W36 — IN3
HV4 - UNUSED — W37 — IN4
HV 5V — W22 — W21 — W22
GND — W27 — W26 — W27

CW-022 - 4 RELAYS, 5 V
leave JD-VCC <-> VCC jumper installed

VCC/JD-VCC — W23 — W23
GND — W28 — W28
K1 COM — W41
K1 NO — W42
K2 COM — W38
K2 NO — W39
K3 COM — W43
K3 NO — W44
K4 COM
K4 NO
K4 - UNUSED

LOADS

K1 - DESK LIGHT
230 V AC - COM + NO

W41 — COM
W42 — NO

12 V PSU
separate supply

+12V — W38
0V — W39

WALL LIGHT
12 V DC

\+ — W39
\- — W40

W40

230 V AC: connect only through COM and NO.
Verify the physical terminal order on the relay board.
Mains wiring must be installed by a qualified electrician.

K3 - CEILING LIGHT
230 V AC - COM + NO

W43 — COM
W44 — NO

COLORS: +5 V · 3.3 V · GND · I2S · GPIO · relay control · load wiring

All low-voltage grounds are common. Connect the speaker only between SPK+ and SPK-.